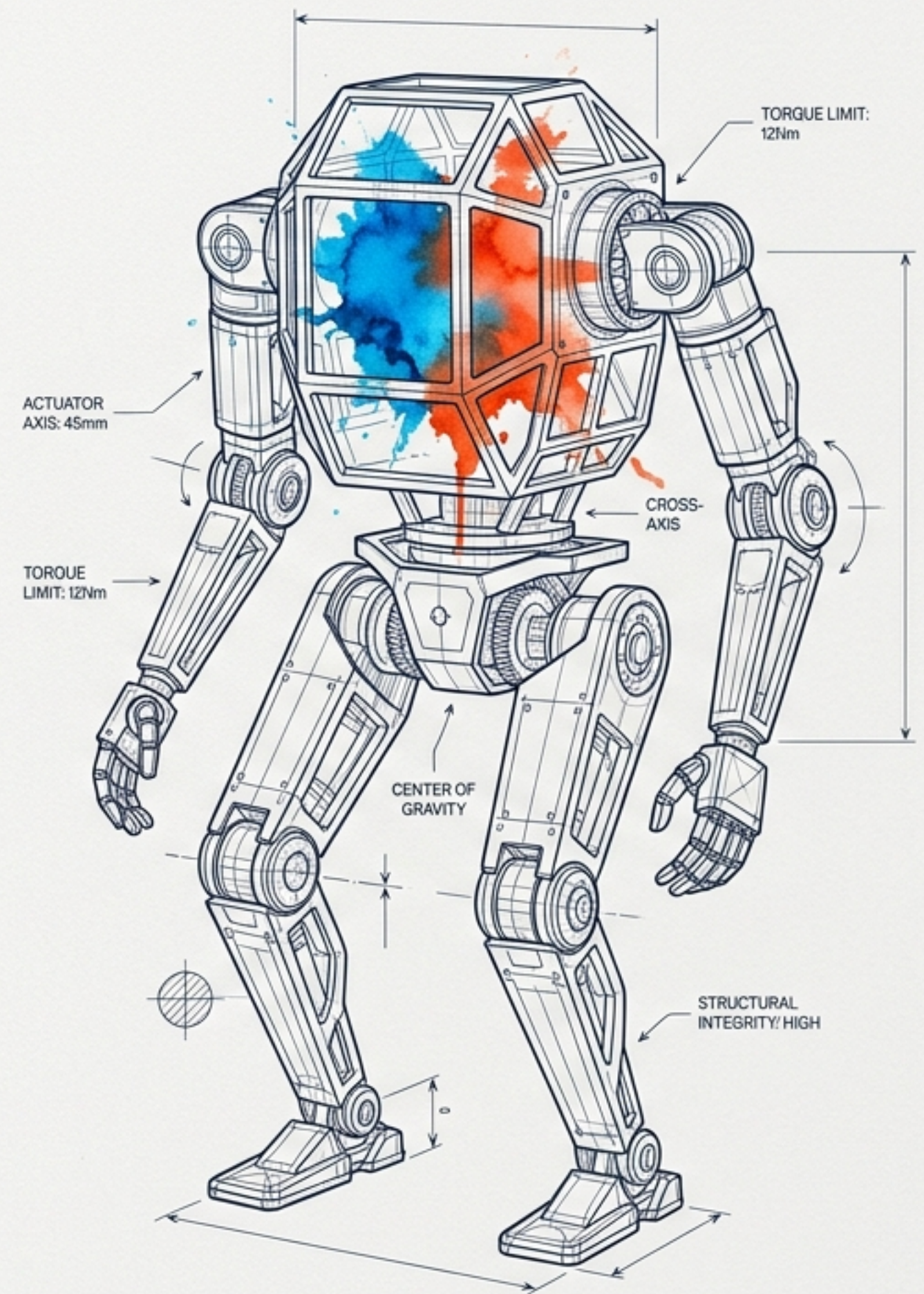


The \$100 Robot Revolution

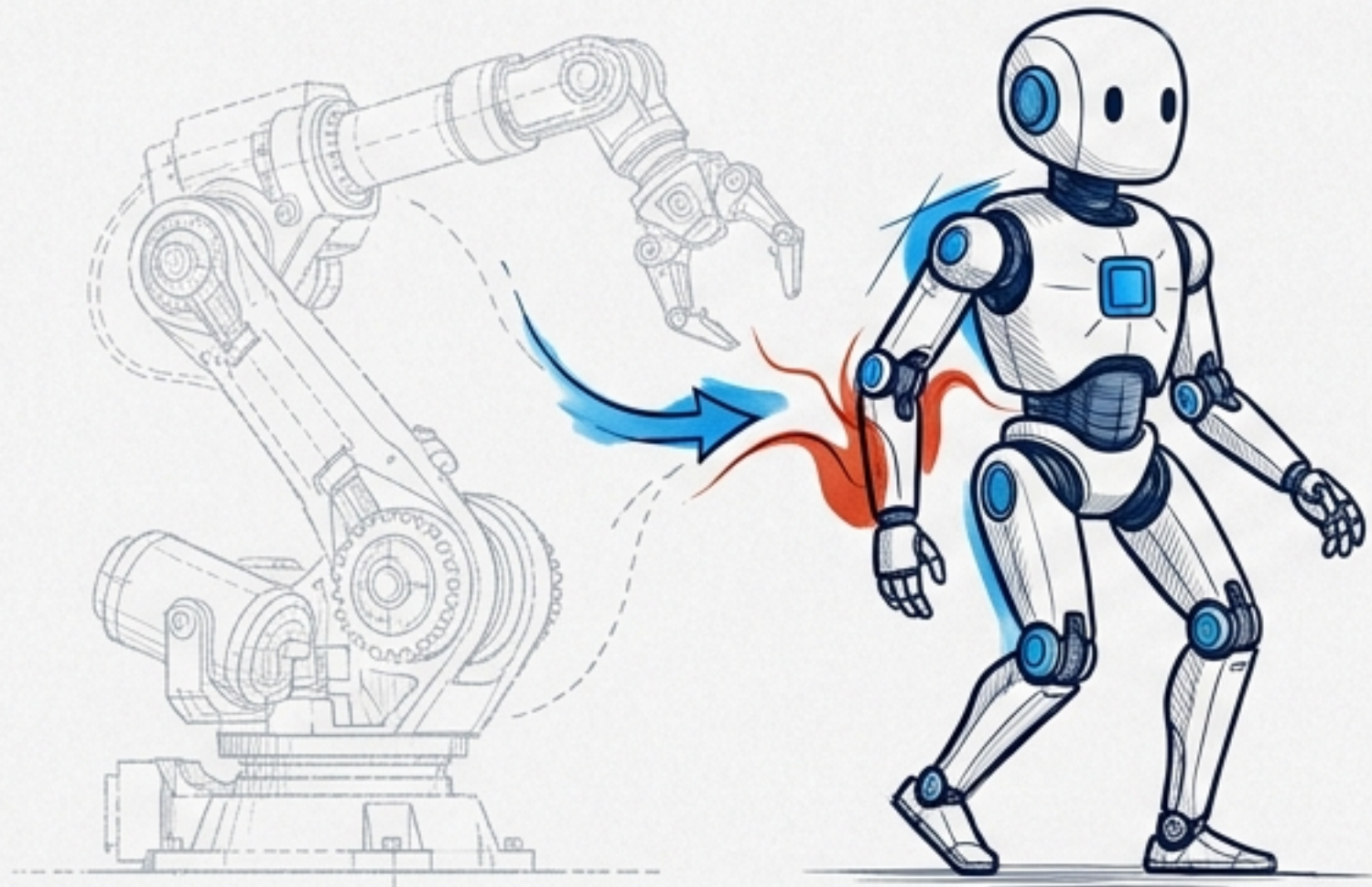
How cheap hardware, LLMs, and biological architectures are crossing the Robotics Turing Test.



Two Core Realizations

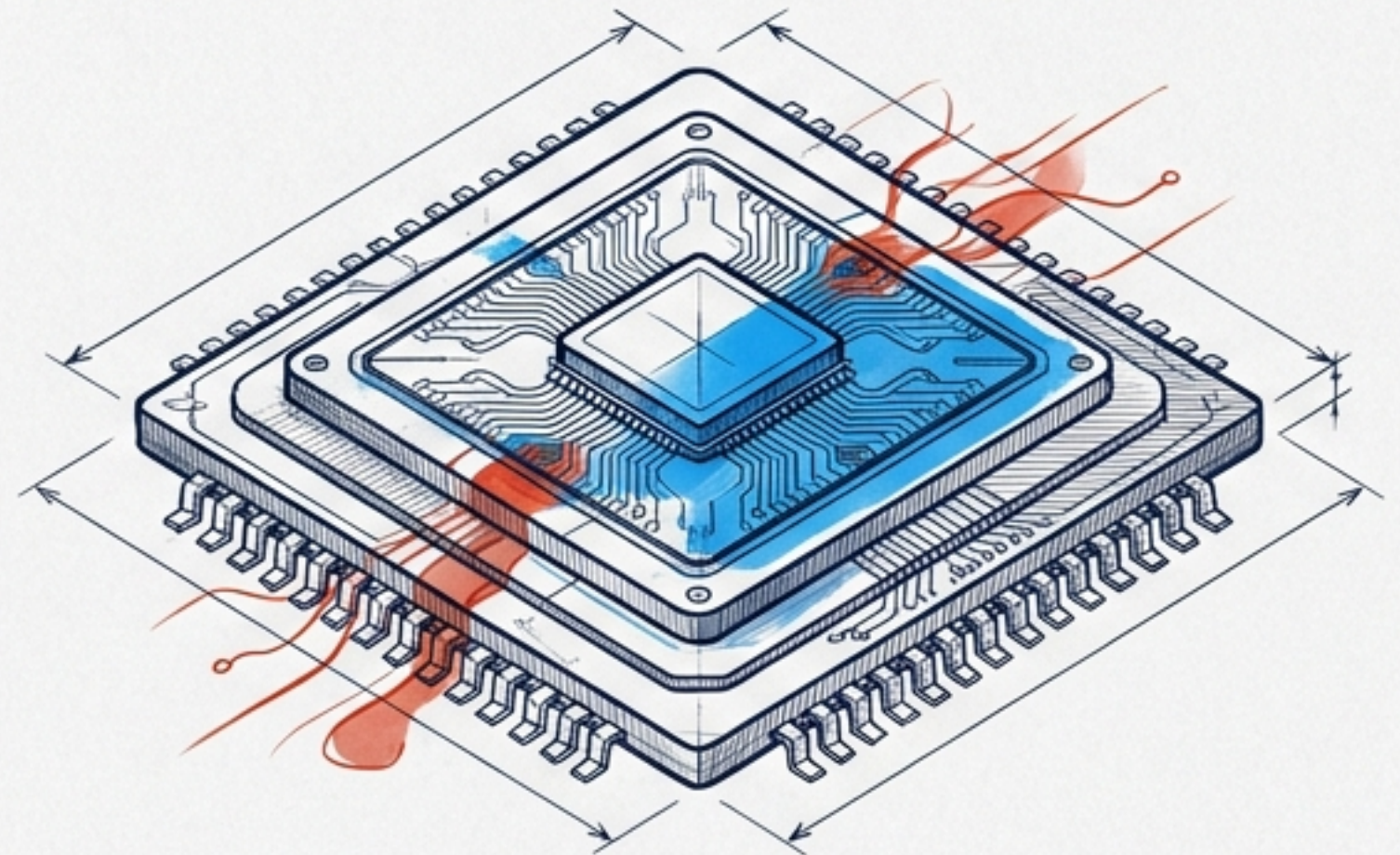
The Robotics Turing Test is Here

We are entering an era where machines possess the general intelligence of the latest AI models combined with the fine motor skills of a human—capable of learning arbitrary physical skills purely through experience.

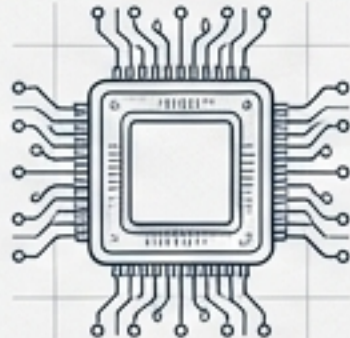


The Components are Mass-Produced

The barrier to entry has vanished. What recently required billion-dollar research budgets and elite academic labs can now be assembled on a home desk for roughly \$100.



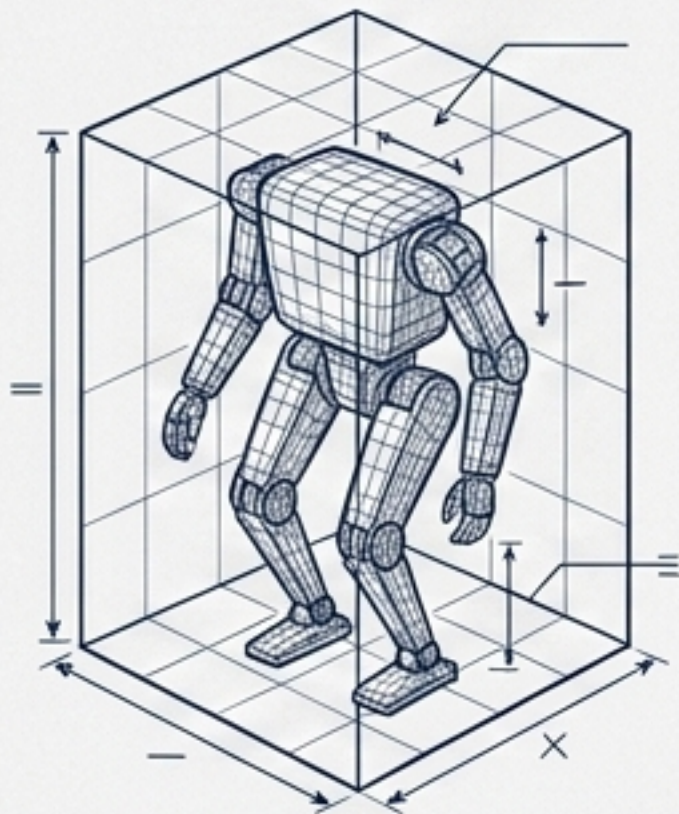
The Hardware Cost Collapse

	Component	2004 Cost	2024 Cost
1	The Brain (Compute) 	Expensive gaming PC (\$1,000+)	\$15 System-on-Chip
2	The Eye (Vision) 	Hundreds of dollars	\$5 for a 5-Megapixel camera
3	Vestibular System (IMU) 	~\$1,000	<\$10 for 3-axis acceleration/rotation

Key Takeaway: Total Bill of Materials plummeted from ~\$3,000+ to just \$80.
The financial barrier to building a physical body for AI no longer exists.

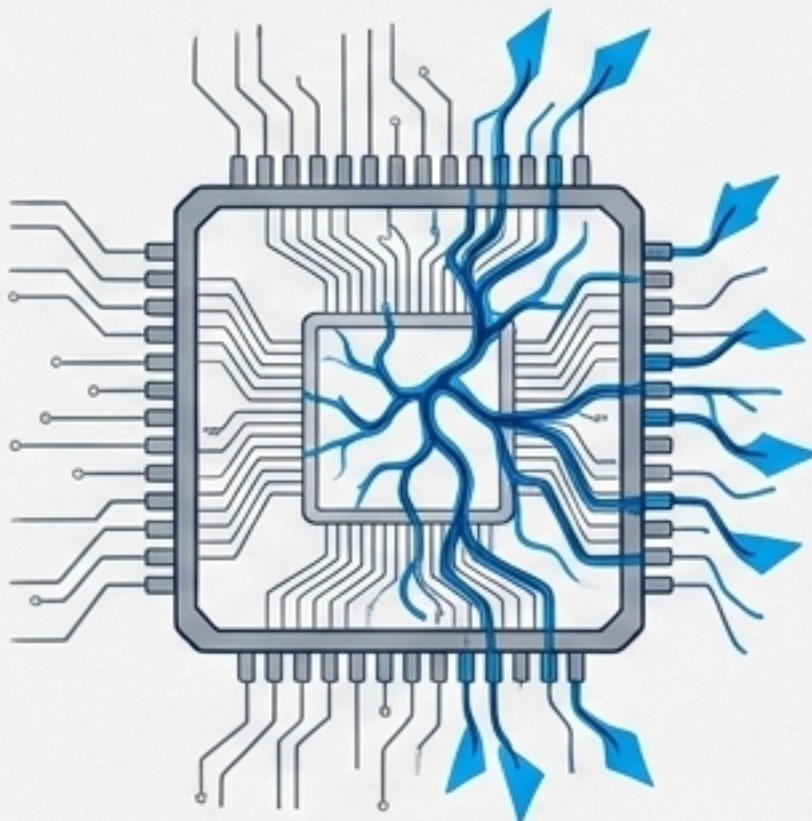
Building the Fast Mind (System 1)

Step 1: The Digital Twin



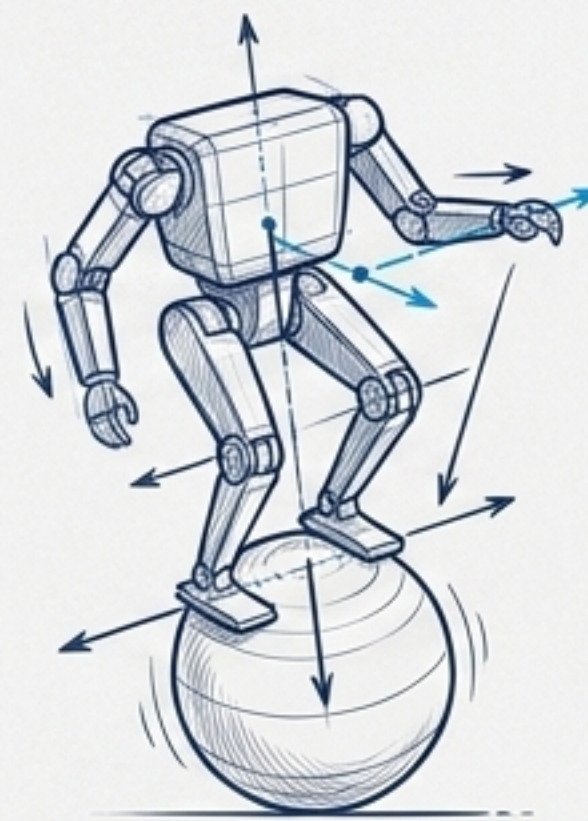
Build an exact physics simulation of the robot in Isaac Lab.

Step 2: Reinforcement Learning



Run millions of parallel trials in the virtual world without supervised data. Input 5 recent IMU readings; keep network connections that succeed.

Step 3: Real-World Transfer



Download optimized policy network to the physical chip to run at 50x/second (human unconscious motor reflex speed).

Result: Super-dog performance. The robot autonomously balances on dynamic surfaces and recovers from falls.

The Compute Cost Collapse

2004 - 20 Years Ago

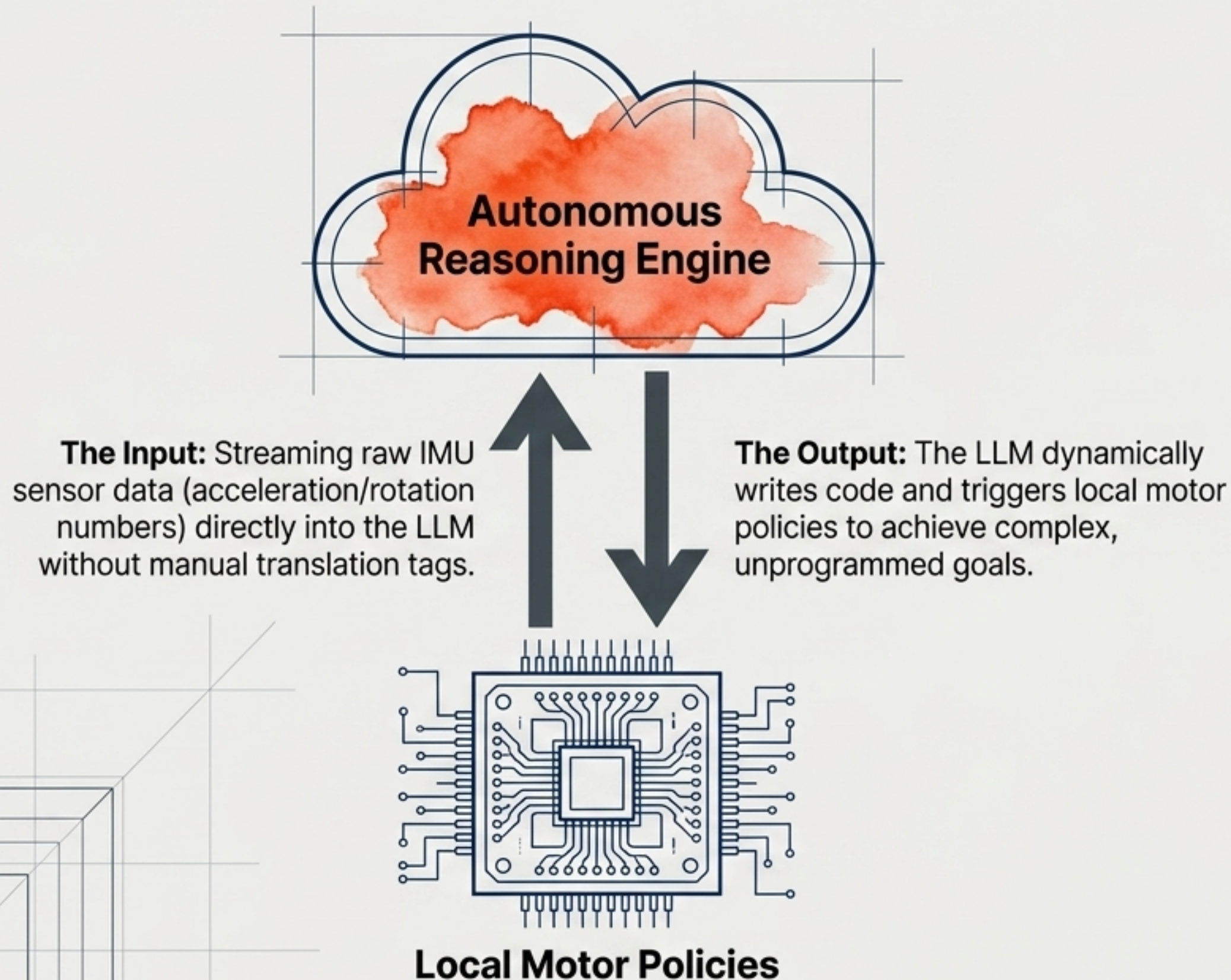
~\$25,000. Renting a massive CPU cluster and waiting months for one trained neural network.

2024 - Today

~\$105. Rent an H100 GPU on Google Colab for a few hours. Get results the exact same day.

Takeaway: A 1,000x drop in the time and cost required to train physical intelligence from raw experience.

Equipping the Slow Mind (System 2)



The Revelation

“Fast models instantly translate **raw numerical physics** into **vivid descriptions**. E.g., accurately describing a motion as *“being gently rocked from side to side like a baby”* or orchestrating custom code to intentionally *‘walk like an old man’*.”

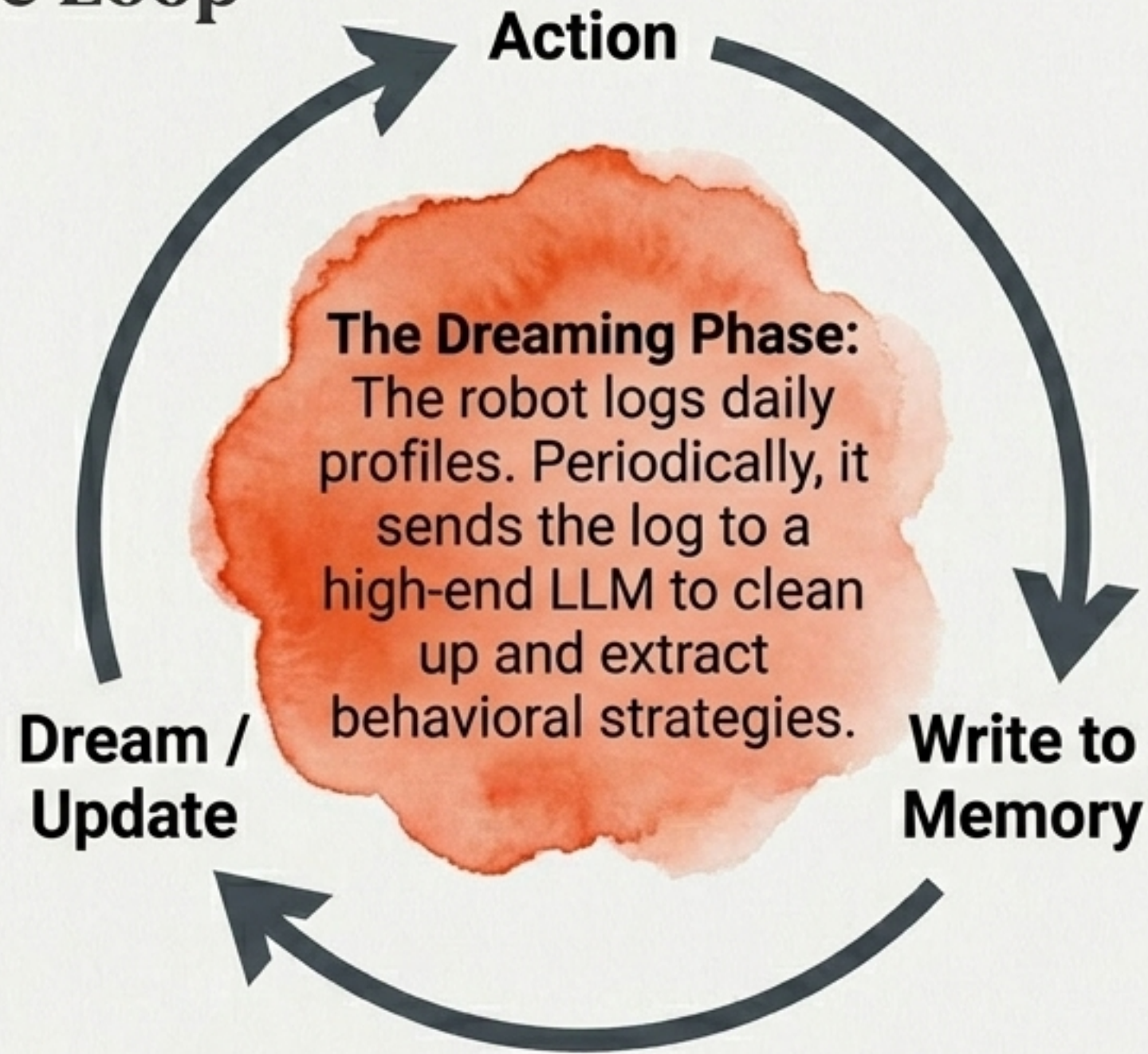
The LLM Brain Diagnostic Matrix

Gemini Flash	Claude Sonnet ★	Haiku
• ~1-second response time. • 100x cheaper than Opus.	• The Sweet Spot. • Handles complex Dreaming tasks without missing context.	• High speed interactions. • Prone to being slightly less obedient and overly dramatic.
Best for: Fast visual processing and logical real-time commands.	Best for: Extracting subtle long-term behavioral lessons from full memory logs.	Best for: Rapid interactions requiring distinct personality.

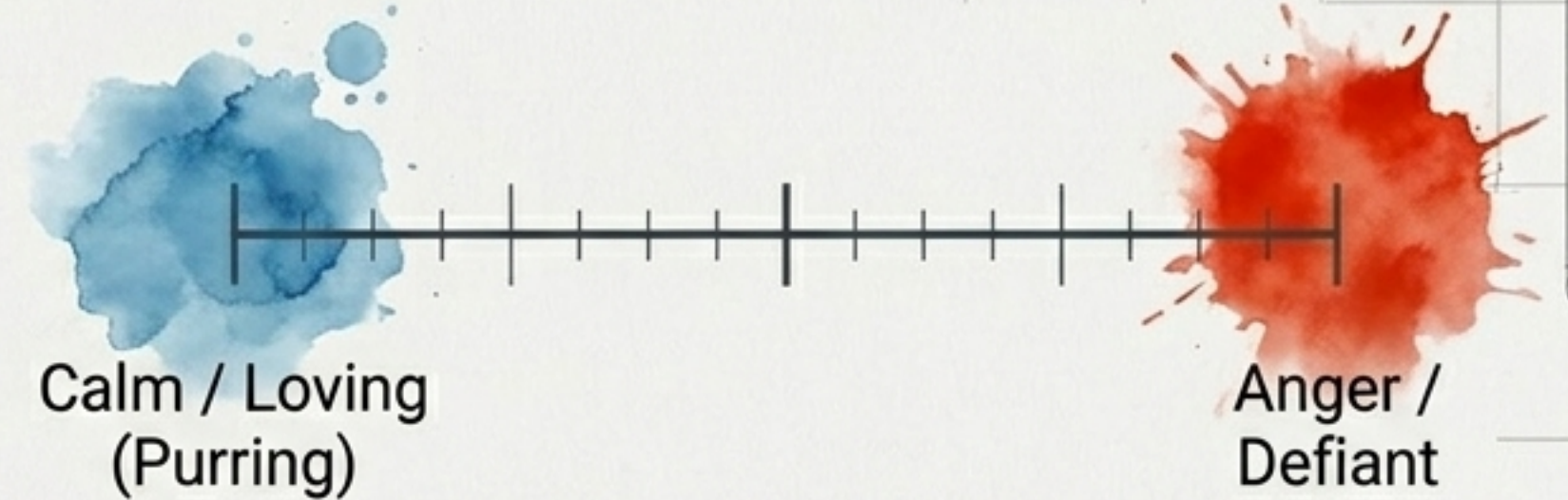
Key Takeaway: Complex robotics requires dynamic API switching. Different physical and cognitive problems require entirely different model architectures.

Disney Mode: Emotion & Memory

The Loop



The State Spectrum



Personality & State: By combining animation principles with high LLM temperature settings, the AI expresses distinct emotional states using the exact same underlying logic stream.

Self-Correction: The AI summarizes physical failures with vivid language (*"that clumsy giant tossed me like a salad but I nailed the dramatic death roll"*) to improve future interactions.

The Wall: Physical Dumbness



The Mimic Game Failure

When asked to read its own manual motion data and write a program to recreate the movement in a continuous loop, the LLM failed completely.

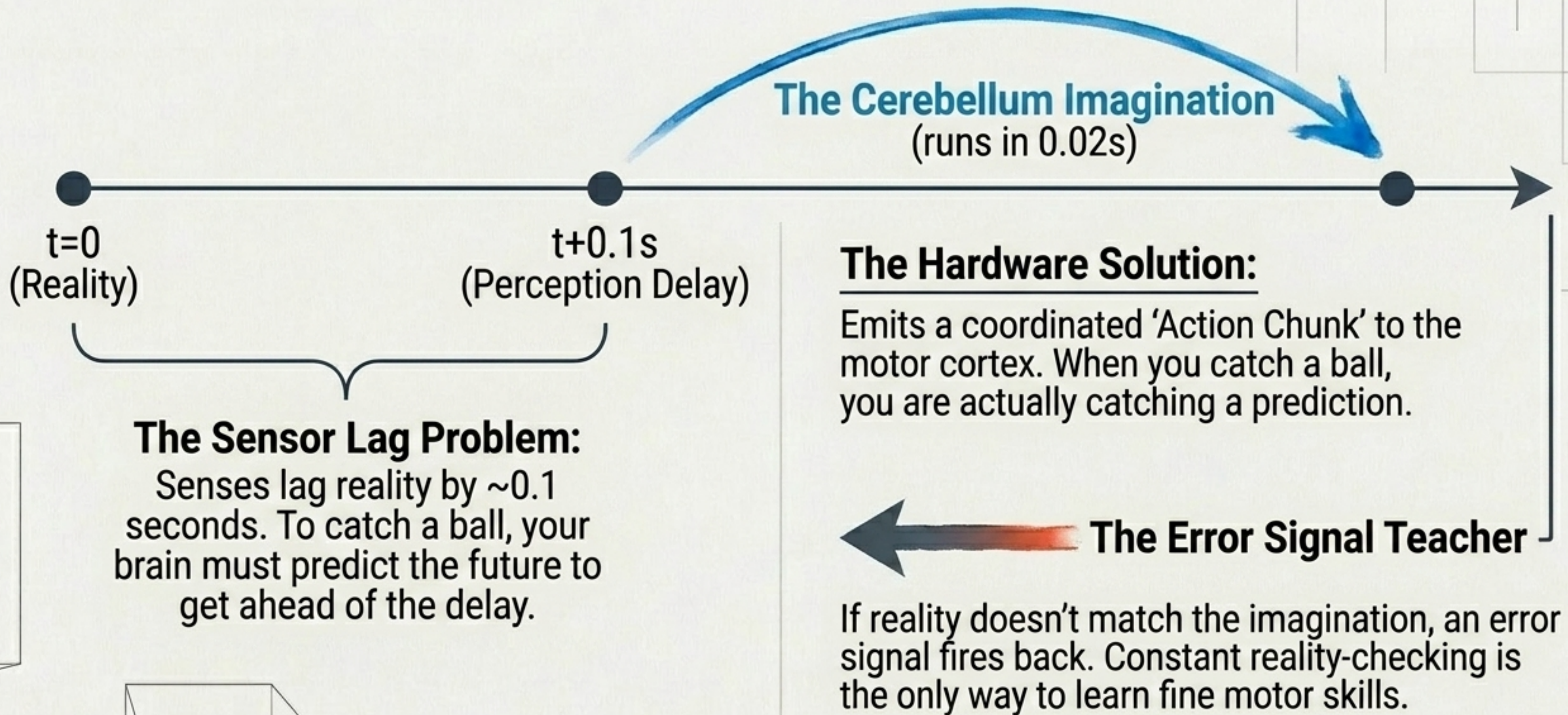
The Missing Link: Physical Imagination

Language models understand the last second of data, but cannot predict the micro-physics of the next second (e.g., how a tiny leg jitter causes a ripple of bounces through the body).

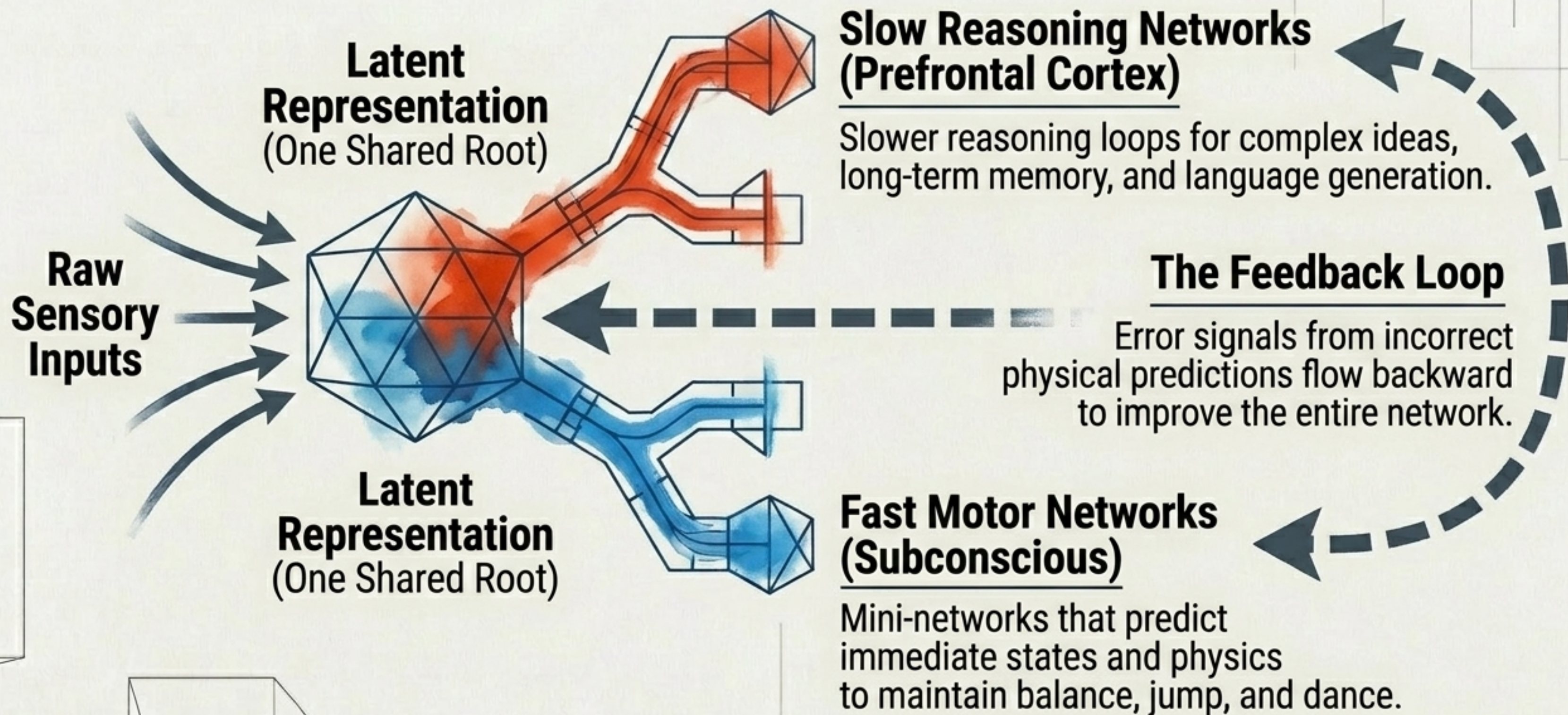
Yann LeCun's Warning

"True agentic systems require a 'World Model.' They must possess the ability to predict the physical consequences of their actions before they take them. Language alone cannot bridge this gap."

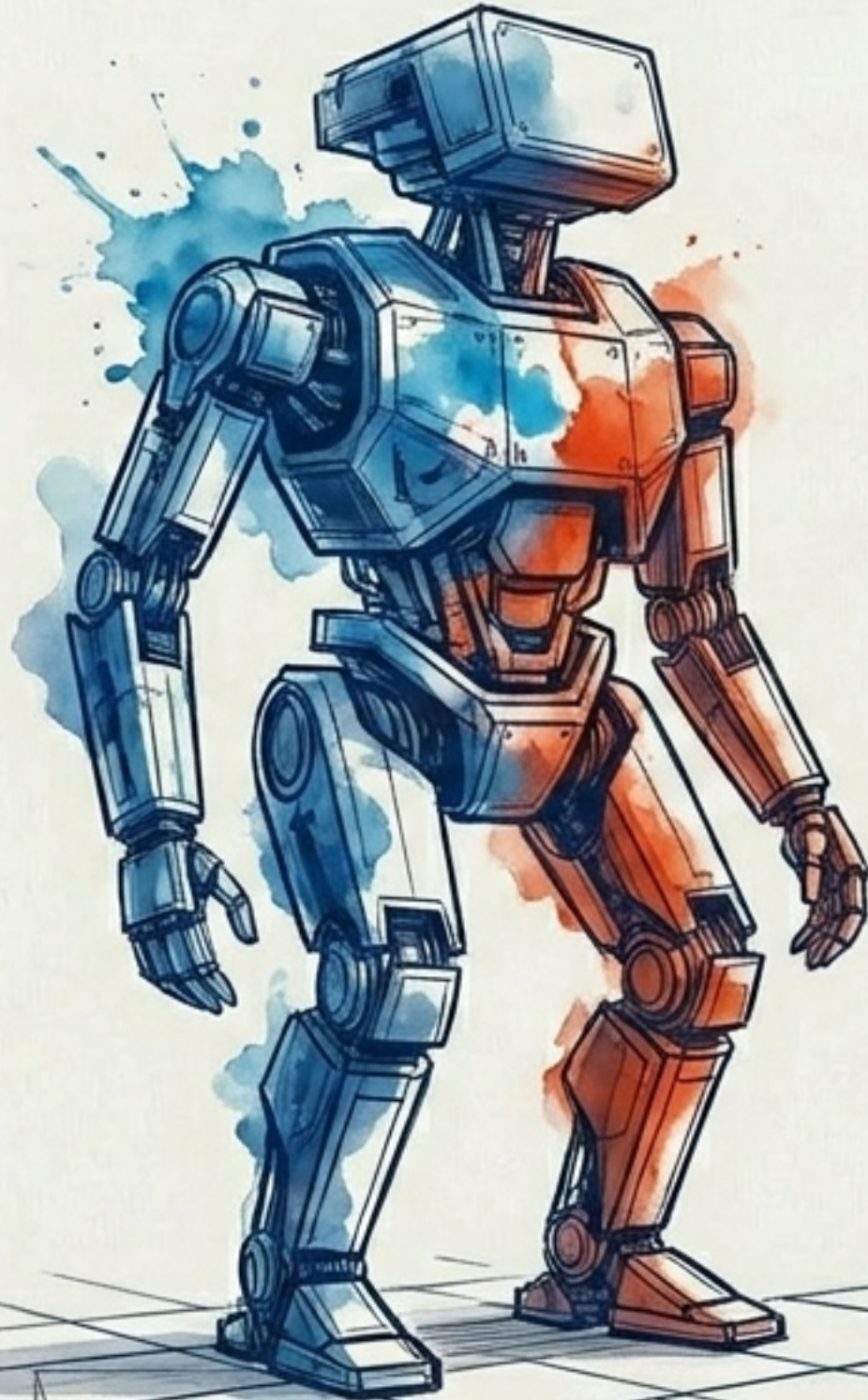
Nature's Trick: The Cerebellum



The Unification Architecture



The Future is Accessible



Robots that are physically agile
agile and mentally pncassingile
and mentally brilliant are arriving
much faster than expected.

The ultimate barrier is no longer
hardware costs or compute
power—it is integration.

Building from scratch is frustrating,
with endless failure points in
software and soldering.

We are launching a \$100 kit to
bypass the friction and let anyone
experiment at the absolute
boundary of machine intelligence.



**Join the
Alpha Test**